

# Maven, Docker and Kubernetes Flow

## 1. MongoDB Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: employeedb
spec:
  replicas: 1
  selector:
    matchLabels:
      app: employeedb
  template:
    metadata:
      labels:
        app: employeedb
    spec:
      containers:
        - name: employeedb
          image: mongo:latest
          ports:
            - containerPort: 27017
```

## 2. MongoDB Service

```
apiVersion: v1
kind: Service
metadata:
  name: employeedb
spec:
  selector:
    app: employeedb
  ports:
    - protocol: TCP
      port: 27017
      targetPort: 27017
  type: ClusterIP
```

## 3. EmployeeService Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: employeeservice
```

## Maven, Docker and Kubernetes Flow

```
spec:
  replicas: 1
  selector:
    matchLabels:
      app: employeeservice
  template:
    metadata:
      labels:
        app: employeeservice
    spec:
      containers:
        - name: employeeservice
          image: employee-service:1.1
          ports:
            - containerPort: 8080
          env:
            - name: SPRING_DATA_MONGODB_URI
              value: mongodb://employeedb:27017/employeeManagement
```

### 4. EmployeeService Service

```
apiVersion: v1
kind: Service
metadata:
  name: employeeservice
spec:
  selector:
    app: employeeservice
  ports:
    - protocol: TCP
      port: 9090
      targetPort: 8080
  type: NodePort
```

### Flow Diagram

```
[Maven Build]
  |
  v
[ Docker Image Created ]
  |
  v
```

## Maven, Docker and Kubernetes Flow

[Kubernetes]

|

v

[MongoDB Deployment + Service] <--> [EmployeeService Deployment + Service]

- MongoDB on port 27017
- EmployeeService on port 9090 (NodePort) --> 8080 (Pod)

### Important Commands

```
kubectl apply -f mongodb-deployment.yaml
kubectl apply -f mongodb-service.yaml
kubectl apply -f employeeservice-deployment.yaml
kubectl apply -f employeeservice-service.yaml
```

### Final Notes

- No need for docker-compose up if you are deploying with Kubernetes.
- Kubernetes handles networking automatically via service names.
- Always build your docker images first before applying deployment files.